

Center, Orbit Decomposition, and Residual Fibers in the Triangular Lattice

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Abstract

We study a center-based reconstruction problem in the equilateral triangular lattice and in centered equilateral boundaries. For the uniform triangular refinement R_n , the geometric center is a lattice vertex exactly when $3 \mid n$; R_3 is the first refinement with a strict interior vertex, and its center has link C_6 . In the infinite A_2 root graph the graph-distance shell S_k has the classical size $6k$. Under the 120° coordinate rotation each shell decomposes into exactly $2k$ disjoint three-point orbits, every one of which has centroid equal to the shell center. For this canonical orbit family, the exact vertex-deletion hitting number is therefore $2k$.

We then define incidence fibers for centered regular boundaries. A general convex-geometric theorem is proved first: for every $n \geq 3$, the $2n$ vertices of a strictly convex C_n -symmetric $2n$ -gon lie on the boundary of exactly two regular n -gons. These two boundaries are obtained from the two alternating perfect matchings of the cycle of polygon edges. For $n = 3$, one embedded C_3 orbit has a one-dimensional circle of centered equilateral boundary realizations. Two alternating radial C_3 contact orbits with ratio $\rho = r_\ell/r_s$ form a strictly convex D_3 -symmetric hexagon for $1 \leq \rho < 2$, so the general theorem gives exactly two equilateral boundaries; the support-function calculation shows that they are congruent and reflection-conjugate. At $\rho = 2$ the hexagon degenerates to a triangle with the short triplet at side midpoints, and the two boundaries coalesce to one; for $\rho > 2$ the short triplet is interior to the long triangle and the incidence fiber is empty. We verify the formulas and the general alternating-edge theorem computationally. Classical A_2 coordination facts and rotationally symmetric enclosure theory are treated as background.

1 Introduction

The graph-distance shells of root lattices and their coordination sequences are classical objects. In particular, coordination sequences for A_2 and other root lattices were summarized and extended by Baake–Grimm [1]; see also Goodman–Strauss–Sloane [2] for a later combinatorial perspective. Accordingly, the shell count $|S_k| = 6k$ is not presented here as a new result.

Our starting point is instead a reconstruction question. The six nearest-neighbor directions around the center of the first nontrivial triangular refinement split naturally into two 120° orbits. The same free order-three action propagates radially: every shell S_k contains $2k$ disjoint rotational orbits, each with the same centroid. We regard these specific group orbits as *canonical orbit certificates*; the associated deletion result is an exact hitting-number statement for this canonical family.

A second question arises when three-point rotational orbits are regarded as boundary contacts rather than lattice vertices. How much of a *centered* enclosing equilateral boundary is determined by one orbit, and how much by two alternating orbits? We formalize this as an incidence fiber in

the two-parameter space of centered equilateral triangles. The answer is a fiber hierarchy: one orbit leaves a one-dimensional circle, while two alternating orbits leave generically two reflection-conjugate boundaries, collapsing to one at the endpoint where the short triplet occupies side midpoints and the long triplet occupies vertices.

There is substantial adjacent literature on enclosing triangles, support geometry, and rotationally symmetric convex bodies. Of particular relevance, Cañete–Miori–Segura Gomis [6] construct a smallest equilateral triangle containing a 3-rotationally symmetric planar convex body and explicitly note that such a triangle need not be unique, with the regular hexagon as a basic example. Cañete–Schnell–Segura Gomis [7] develop the analogous k -rotationally symmetric setting. Minimal enclosing triangles were also studied algorithmically by O’Rourke–Aggarwal–Maddila–Baldwin [8], and fixed-angle minimum-area enclosing triangles by Bose–De Carufel [9]. Chen–Lourie [5] study tripod configurations of curves and related regular-polygon support geometry. These works are optimization, partition, or existence theories; the object below is instead the complete vertex-incidence fiber of a prescribed rotationally symmetric polygon. The distinction is narrow and is stated explicitly to avoid a priority claim for the surrounding support geometry.

The paper is deliberately geometric. No dynamical, physical, psychological, or semantic interpretation is assumed. We also distinguish carefully between an abstract graph cycle C_6 , a six-direction ray frame, the fixed coordinate action $\langle R \rangle$, and embedded three-point rotational orbits.

Novelty scope. The individual facts $|S_k| = 6k$, group averaging under a 120° rotation, elementary support-line formulas, and the existence of regular enclosing polygons for rotationally symmetric convex bodies are standard or immediate. The claims isolated here concern their exact organization into canonical reconstruction certificates, the corresponding hitting number, the alternating-edge incidence theorem for strictly convex C_n -symmetric $2n$ -gons, and the exact D_3 radial specialization with its reflection pair and $2 \rightarrow 1$ degeneration. We make no priority claim for the standard constituent lemmas or for general enclosing-polygon theory.

2 Finite triangular refinements and the first center

Let

$$\Delta = \{(x_1, x_2, x_3) \in \mathbb{R}_{\geq 0}^3 : x_1 + x_2 + x_3 = 1\}$$

be an equilateral triangle in barycentric coordinates. Its uniform refinement R_n has vertex set

$$V(R_n) = \left\{ \frac{1}{n}(i, j, k) : i, j, k \in \mathbb{Z}_{\geq 0}, i + j + k = n \right\}.$$

Two vertices are adjacent when their integer triples differ by a permutation of $(1, -1, 0)$.

The geometric center is

$$c = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

Proposition 2.1. *The center c is a vertex of R_n if and only if $3 \mid n$. The refinement R_3 is the first one with a strict interior lattice vertex, and that vertex is unique. In R_3 ,*

$$\deg(c) = 6, \quad \text{Link}_{R_3}(c) \cong C_6.$$

Proof. The center is a lattice vertex exactly when $n/3$ is an integer. A strict interior vertex corresponds to an integer triple (i, j, k) with all coordinates positive and sum n . Such a triple does not exist for $n < 3$, while for $n = 3$ the unique possibility is $(1, 1, 1)$. The six allowed root steps obtained from permutations of $(1, -1, 0)$ remain inside R_3 at this vertex and form a cyclic link. $\square$

Remark 2.2. For R_{3m} with $m > 1$, the same normalized center persists as $(m, m, m)/(3m)$, but it is no longer the unique interior vertex. Thus R_3 is the first appearance of the center as an interior lattice vertex, not the only refinement containing it.

3 The infinite A_2 shell and its C_3 orbit decomposition

Use the standard root-lattice model

$$A_2 = \{(x, y, z) \in \mathbb{Z}^3 : x + y + z = 0\},$$

with adjacency by the six roots obtained from permutations of $(1, -1, 0)$. The graph distance from the origin is

$$d(0, (x, y, z)) = \frac{|x| + |y| + |z|}{2}. \quad (1)$$

Let

$$S_k := \{v \in A_2 : d(0, v) = k\}.$$

The standard coordination sequence gives $|S_k| = 6k$ for $k \geq 1$ [1]. Geometrically, S_k is the hexagonal graph-distance boundary and the induced graph is C_{6k} .

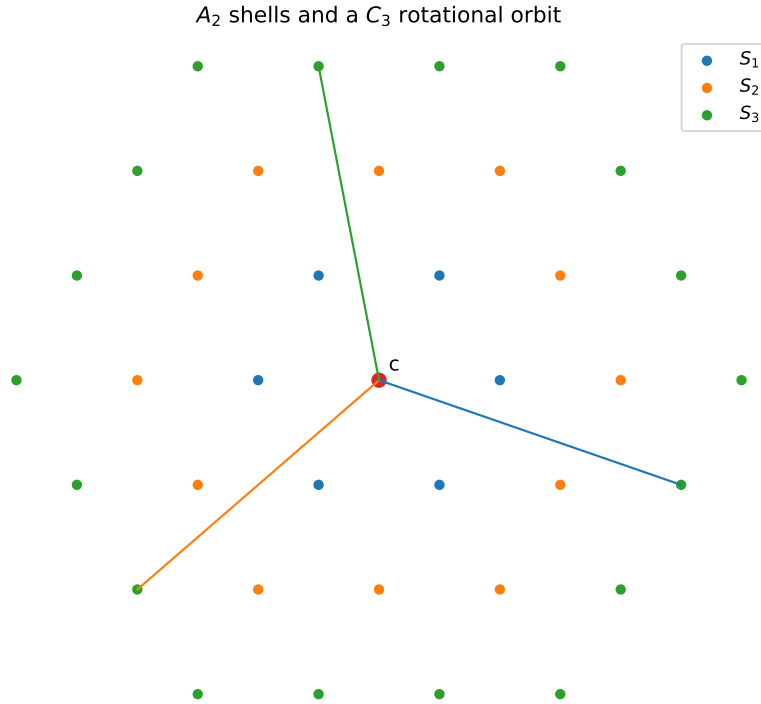


Figure 1: The first three graph-distance shells of A_2 . A sample 120° orbit is joined to the center.

Let R denote the order-three coordinate rotation

$$R(x, y, z) = (y, z, x).$$

It preserves A_2 , adjacency, graph distance, and each shell S_k .

Theorem 3.1 (Rotational orbit decomposition). *For every $k \geq 1$, the shell S_k decomposes into exactly $2k$ disjoint orbits of the cyclic group $\langle R \rangle \cong C_3$:*

$$S_k = \bigsqcup_{j=1}^{2k} \mathcal{O}_j, \quad |\mathcal{O}_j| = 3.$$

For every orbit,

$$\frac{1}{3} \sum_{v \in \mathcal{O}_j} v = 0.$$

Thus every orbit independently reconstructs the shell center by its centroid.

Proof. A fixed point of R satisfies $x = y = z$; together with $x + y + z = 0$, this forces $x = y = z = 0$. Hence R acts freely on every nonzero shell and every orbit has cardinality 3. Since $|S_k| = 6k$, there are $2k$ orbits.

For $v = (x, y, z)$,

$$v + Rv + R^2v = (x + y + z, x + y + z, x + y + z) = (0, 0, 0).$$

Therefore the centroid of each orbit is the origin. $\square$

Definition 3.2 (Canonical orbit certificates). *For a fixed shell S_k and the fixed coordinate rotation $R(x, y, z) = (y, z, x)$, let*

$$\mathfrak{O}_k := S_k / \langle R \rangle$$

be the family of its C_3 orbits. An element of $\mathfrak{O}_k$ is called a canonical orbit certificate. Its centroid reconstructs the shell center by Theorem 3.1.

Corollary 3.3 (Exact hitting number for canonical certificates). *The transversal number of the canonical orbit family is*

$$\tau(\mathfrak{O}_k) = 2k.$$

Equivalently, the minimum number of shell vertices whose deletion intersects every canonical orbit certificate is exactly $2k$. Consequently, after any adversarial deletion of at most $2k - 1$ shell vertices, at least one intact canonical certificate remains and reconstructs the center by averaging.

Proof. The $2k$ members of $\mathfrak{O}_k$ are disjoint. Every hitting set must therefore contain at least one vertex from each orbit, so its size is at least $2k$. Choosing one vertex from each orbit attains this bound. $\square$

Remark 3.4. *The statement is deliberately relative to the canonical group-orbit family $\mathfrak{O}_k$. It does not assert that every three-point subset of S_k whose centroid is the origin must be one of these orbits.*

3.1 Finite truncation in R_{3m}

The infinite-shell result has a simple finite counterpart.

Proposition 3.5. *In R_{3m} , center the integer coordinates at (m, m, m) . For every radius $1 \leq r \leq m$, the graph-distance shell about the center is a complete A_2 shell C_{6r} . The first boundary contact occurs at $r = m$; that shell contains exactly*

$$3(m + 1)$$

vertices of the outer boundary. The shell $r = m + 1$ is truncated.

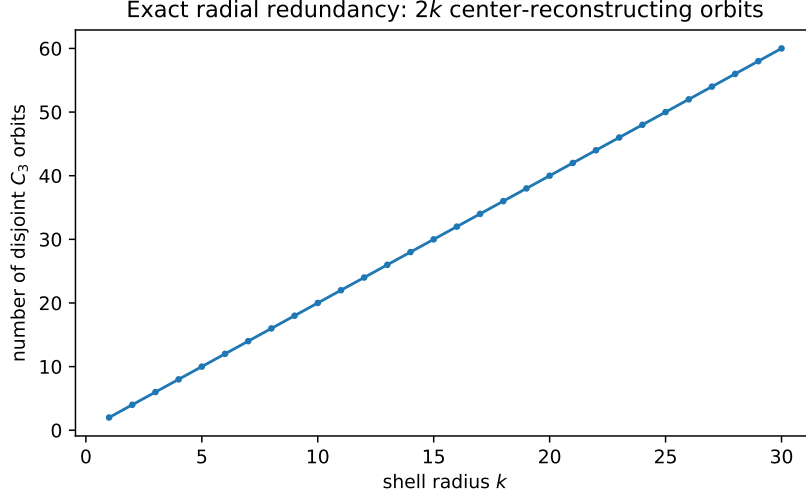


Figure 2: The number of disjoint three-point center-reconstruction orbits grows exactly as $2k$.

Proof. Any A_2 displacement of graph distance r has each coordinate at least $-r$. Thus for $r \leq m$, adding the displacement to (m, m, m) keeps all coordinates nonnegative, so the full shell is present. At $r = m$, boundary points are those with one coordinate equal to zero. On, say, the side $x = 0$, the remaining coordinates sum to $3m$; distance m from (m, m, m) forces both to lie between m and $2m$, giving $m + 1$ points. The three sides give $3(m + 1)$ points and these sets do not overlap at outer vertices. At radius $m + 1$, an infinite-lattice shell contains points with one displacement coordinate $-(m + 1)$, which cannot occur in the finite triangle. $\square$

4 Embedded three-point orbits

We now pass to the Euclidean plane and distinguish group orbits from abstract graph cycles. Let

$$u(\theta) = (\cos \theta, \sin \theta).$$

For a center $c \in \mathbb{R}^2$, phase ψ , and radius $r > 0$, define the embedded rotational triplet

$$\mathcal{T}(c, \psi, r) := \{c + r u(\psi + 2\pi j/3) : j = 0, 1, 2\}. \quad (2)$$

Its centroid is immediately c .

As an embedded object, a triplet has four continuous parameters:

$$(c_x, c_y, \psi, r).$$

Two alternating triplets of a common six-ray frame take the form

$$\mathcal{T}_+ = \mathcal{T}(c, \psi, r_+), \quad (3)$$

$$\mathcal{T}_- = \mathcal{T}(c, \psi + \pi/3, r_-), \quad (4)$$

and therefore have five joint continuous parameters $(c_x, c_y, \psi, r_+, r_-)$. In this differential sense, the two separate four-parameter descriptions share three parameters: the two center coordinates and the common phase relation.

5 Boundary fibers from one triplet

For fixed $c \in \mathbb{R}^2$, define the parameter space of centered equilateral triangles

$$\mathcal{B}_c := (\mathbb{R}/(2\pi/3)\mathbb{Z}) \times \mathbb{R}_{>0}.$$

A parameter pair $(\beta, h) \in \mathcal{B}_c$ defines the closed triangle

$$B(c; \beta, h) := \bigcap_{j=0}^2 \{x \in \mathbb{R}^2 : u(\beta + 2\pi j/3) \cdot (x - c) \leq h\}. \quad (5)$$

Its inradius is h , its side-normal phase is β modulo $2\pi/3$, and its side length is $L = 2\sqrt{3}h$. For a finite point set $X \subset \mathbb{R}^2$, define the centered boundary-incidence fiber

$$\mathcal{F}_c(X) := \{(\beta, h) \in \mathcal{B}_c : X \subseteq \partial B(c; \beta, h)\}. \quad (6)$$

For $\theta \in \mathbb{R}$, let

$$d_3(\theta) := \min_{m \in \mathbb{Z}} |\theta - 2\pi m/3| \in [0, \pi/3], \quad (7)$$

where the representative is chosen after reduction modulo $2\pi/3$.

Lemma 5.1 (Support value of a rotational triplet). *For $\mathcal{T} = \mathcal{T}(c, \psi, r)$ and $(\beta, h) \in \mathcal{B}_c$,*

$$\mathcal{T} \subseteq \partial B(c; \beta, h) \iff \boxed{h = r \cos d_3(\beta - \psi)}.$$

Proof. For any triplet point $x = c + ru(\psi + 2\pi m/3)$, the largest of its three support values in (5) is

$$r \max_{j=0,1,2} \cos(\psi - \beta + 2\pi(m - j)/3) = r \cos d_3(\beta - \psi).$$

The point is in the triangle exactly when this maximum is at most h , and it lies on the boundary exactly when equality holds. The value is the same for all three points because their phases differ by $2\pi/3$. $\square$

Theorem 5.2 (One-triplet boundary fiber). *For a fixed embedded triplet $\mathcal{T}(c, \psi, r)$,*

$$\boxed{\mathcal{F}_c(\mathcal{T}) = \{(\beta, h) : h = r \cos d_3(\beta - \psi)\}}. \quad (8)$$

Equivalently, with the wrapped offset $\delta \in [-\pi/3, \pi/3]$,

$$\beta = \psi + \delta \pmod{2\pi/3}, \quad h = r \cos \delta, \quad L(\delta) = 2\sqrt{3}r \cos \delta.$$

The endpoints $\delta = -\pi/3$ and $\delta = \pi/3$ represent the same element of $\mathcal{B}_c$. Hence the incidence fiber is a one-dimensional circle,

$$\boxed{\mathcal{F}_c(\mathcal{T}) \cong S^1}.$$

In particular, one complete embedded C_3 orbit determines its center exactly but not its centered equilateral boundary.

Proof. The formula is exactly Theorem 5.1. As β runs once through $\mathbb{R}/(2\pi/3)\mathbb{Z}$, the corresponding positive h is uniquely determined and varies continuously between $r/2$ and r . Thus the graph of this continuous function over the phase circle is itself homeomorphic to S^1 . $\square$

One embedded C_3 orbit leaves a one-parameter boundary fiber

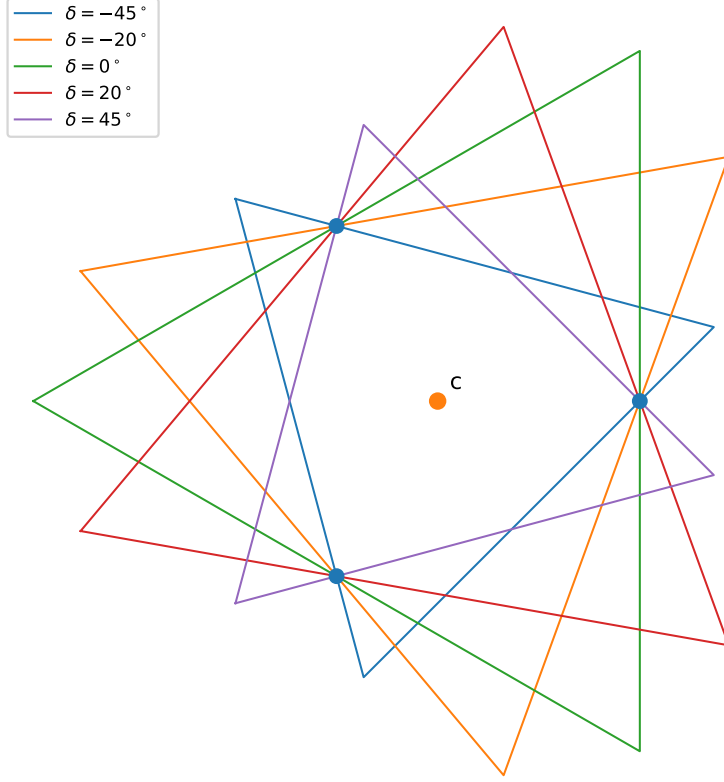


Figure 3: Several members of the one-parameter family in Theorem 5.2, all passing through the same embedded three-point orbit.

6 Alternating-edge incidence for rotationally symmetric polygons

The exact two-boundary phenomenon is not special to the trigonometric parametrization used below. It is a consequence of strict convexity together with an order- n rotational action on a $2n$ -cycle.

Definition 6.1. Let $P = \text{conv}\{v_0, \dots, v_{2n-1}\}$ be a strictly convex polygon with vertices in cyclic order. Assume that P is invariant under the rotation R of angle $2\pi/n$ about a point c , and that the labeling is chosen so that

$$Rv_i = v_{i+2} \pmod{2n}.$$

Let $e_i = [v_i, v_{i+1}]$ denote its edges. Define $\mathcal{G}_n(P)$ to be the family of regular n -gons Q such that

$$P \subseteq Q, \quad \{v_0, \dots, v_{2n-1}\} \subseteq \partial Q.$$

Theorem 6.2 (Alternating-edge boundary theorem). For every $n \geq 3$ and every strictly convex C_n -symmetric $2n$ -gon P as above,

$$|\mathcal{G}_n(P)| = 2.$$

The two elements are obtained by taking the supporting lines of the even edges

$$e_0, e_2, \dots, e_{2n-2}$$

and of the odd edges

$$e_1, e_3, \dots, e_{2n-1},$$

respectively. Both regular n -gons are centered at the rotational center c .

Proof. Let $Q \in \mathcal{G}_n(P)$. Since Q is convex and contains P , each side line of Q that meets a vertex of P is a supporting line of P . Strict convexity implies that a supporting line of P contains at most two vertices, and if it contains two then they are adjacent and determine one edge e_i .

There are n side lines of Q and $2n$ distinct vertices of P to be covered. Hence every side must contain exactly two vertices. No vertex can lie at a vertex of Q , because then it would belong to two side lines and the union of the n side-line vertex sets would contain fewer than $2n$ distinct vertices. Consequently the n selected edges of P form a perfect matching of the cycle C_{2n} . A cycle of even length has exactly two perfect matchings by edges: the even matching and the odd matching displayed above.

Conversely, the even edges form one orbit under R , and so do the odd edges. The supporting line of e_{i+2} is the image under the rotation R of the supporting line of e_i . Thus each alternating family consists of n supporting lines with equal distance from c and successive normal directions differing by $2\pi/n$. Their inward half-planes intersect in a regular n -gon centered at c which contains P . Because the selected edges form a perfect matching, every vertex of P lies on one of its sides. Hence both matchings produce members of $\mathcal{G}_n(P)$, and the preceding argument shows that there are no others. The two resulting polygons are distinct: if their side-line sets coincided, some supporting line of the strictly convex polygon P would contain both an even edge and an odd edge. A supporting line of a strictly convex polygon contains at most one edge, so this is impossible. $\square$

Remark 6.3. For a regular $2n$ -gon, extending alternating sides to obtain two regular n -gons is elementary. Theorem 6.2 uses only the weaker hypotheses of strict convexity and C_n symmetry. It is an incidence statement, not a minimum-area or minimum-inradius assertion.

7 Two alternating triplets and the residual two-point fiber

Let

$$\mathcal{T}_s = \mathcal{T}(c, \psi, r_s), \tag{9}$$

$$\mathcal{T}_\ell = \mathcal{T}(c, \psi + \pi/3, r_\ell), \tag{10}$$

with $0 < r_s \leq r_\ell$ and

$$\rho := \frac{r_\ell}{r_s} \geq 1.$$

Let $H_\rho := \text{conv}(\mathcal{T}_s \cup \mathcal{T}_\ell)$.

Proposition 7.1 (Convex-hull transition). *The six contacts have the following exact convex-hull behavior.*

- (i) If $1 \leq \rho < 2$, then H_ρ is a strictly convex D_3 -symmetric equilateral hexagon. Every edge has length

$$r_s \sqrt{\rho^2 - \rho + 1},$$

and every edge line has distance

$$h_\rho = \frac{\sqrt{3} \rho r_s}{2\sqrt{\rho^2 - \rho + 1}} \tag{11}$$

from c .

- (ii) If $\rho = 2$, the long-radius triplet is the vertex set of an equilateral triangle and the short-radius triplet consists of its three side midpoints.
- (iii) If $\rho > 2$, the short-radius triplet lies strictly in the interior of the triangle spanned by the long-radius triplet.

Proof. After translation, rotation, and scaling, take a short point at $(1, 0)$ and its two neighboring long points at $\rho u(\pm\pi/3)$. The chord joining those long points is the line $x = \rho/2$. Hence the short point is an extreme vertex exactly when $\rho < 2$, lies on the chord at $\rho = 2$, and lies inside the long triangle when $\rho > 2$. Rotational symmetry gives the same conclusion at the other two short rays.

For $\rho < 2$, every polygon edge joins one short and one long point separated by $\pi/3$, so the cosine law gives the displayed common edge length. The distance from the origin to such a chord is its doubled-triangle-area divided by its length:

$$\frac{r_s r_\ell \sin(\pi/3)}{r_s \sqrt{\rho^2 - \rho + 1}} = \frac{\sqrt{3} \rho r_s}{2 \sqrt{\rho^2 - \rho + 1}}.$$

The configuration also has the reflections through the three short-radius axes, giving D_3 symmetry. $\square$

For a candidate boundary phase β , define the two reduced offsets

$$\delta_s := d_3(\beta - \psi), \quad \delta_\ell := d_3(\beta - \psi - \pi/3). \quad (12)$$

Because the two phase classes are separated by $\pi/3$ on the circle $\mathbb{R}/(2\pi/3)\mathbb{Z}$,

$$\boxed{\delta_s + \delta_\ell = \pi/3.} \quad (13)$$

If both triplets lie on the same boundary, Theorem 5.1 gives

$$h = r_s \cos \delta_s = r_\ell \cos \delta_\ell. \quad (14)$$

Since $r_\ell \geq r_s$, equation (14) implies $\cos \delta_s \geq \cos \delta_\ell$, hence $\delta_s \leq \delta_\ell$. Therefore

$$0 \leq \delta_s \leq \pi/6.$$

Writing $\delta := \delta_s$ and using (13), we obtain

$$\boxed{\rho = \frac{\cos \delta}{\cos(\pi/3 - \delta)}, \quad \tan \delta = \frac{2 - \rho}{\sqrt{3} \rho}.} \quad (15)$$

The map $\delta \mapsto \rho$ decreases bijectively from 2 to 1 on $[0, \pi/6]$.

Theorem 7.2 (Two-triplet boundary fiber). *Let $X = \mathcal{T}_s \cup \mathcal{T}_\ell$ be the two alternating triplets above.*

- (i) *The centered incidence fiber $\mathcal{F}_c(X)$ is nonempty if and only if*

$$\boxed{1 \leq \rho \leq 2.}$$

- (ii) *For $1 \leq \rho < 2$, the fiber has exactly two elements. If $\delta \in (0, \pi/6]$ is the unique solution of (15), their phases are*

$$\boxed{\beta_\pm = \psi \pm \delta \pmod{2\pi/3}.}$$

Both have

$$\boxed{h = \frac{\sqrt{3} \rho r_s}{2 \sqrt{\rho^2 - \rho + 1}}, \quad L = \frac{3 \rho r_s}{\sqrt{\rho^2 - \rho + 1}}.} \quad (16)$$

(iii) The two boundaries are exchanged by reflection about the axis of phase ψ :

$$\boxed{\beta_- \equiv 2\psi - \beta_+ \pmod{2\pi/3}.}$$

(iv) At the fiber-collapse endpoint $\rho = 2$, one has $\delta = 0$ and

$$\boxed{|\mathcal{F}_c(X)| = 1.}$$

In that boundary the short-radius triplet occupies the three side midpoints and the long-radius triplet occupies the three vertices.

Proof. For $1 \leq \rho < 2$, Theorem 7.1 makes the six points the vertices of a strictly convex C_3 -symmetric hexagon. Hence Theorem 6.2 already gives exactly two equilateral incidence boundaries. The support-function equations identify them explicitly. Every common boundary satisfies (13)–(15); the displayed ratio takes exactly the values $[1, 2]$, and for $\rho < 2$ its unique solution has $0 < \delta \leq \pi/6$. Conversely, either phase $\beta = \psi \pm \delta$ satisfies both support equalities, so Theorem 5.1 puts all six points on the boundary. The equation $d_3(\beta - \psi) = \delta$ has exactly these two solutions modulo $2\pi/3$.

At $\rho = 2$, Theorem 7.1 shows directly that the short points are side midpoints of the long equilateral triangle, so the fiber has one element. If $\rho > 2$, the short triplet lies strictly inside the convex hull of the long triplet and therefore cannot lie on the boundary of any convex triangle containing all six points. Thus the fiber is empty outside $[1, 2]$. Formula (16) follows by substituting $\tan \delta = (2 - \rho)/(\sqrt{3}\rho)$ into $h = r_s \cos \delta$ and using $L = 2\sqrt{3}h$.

At $\rho = 2$, $h = r_s$ and $\beta = \psi$. The short triplet lies in the three side-normal directions at distance equal to the inradius, hence at the side midpoints. The long triplet has radius $2h$ and phase offset $\pi/3$, which is exactly the vertex radius and vertex phase of the equilateral triangle. $\square$

Corollary 7.3 (Query-dependent reconstruction). *Within the alternating-triplet model class:*

- (a) *either complete triplet alone recovers c by its centroid;*
- (b) *the union $\mathcal{T}_s \cup \mathcal{T}_\ell$ determines the radial ratio $\rho = r_\ell/r_s$, whereas either triplet alone leaves the other radius, and hence ρ , unspecified unless supplied externally;*
- (c) *for $1 \leq \rho < 2$, the union determines the centered enclosing boundary up to exactly one reflection choice;*
- (d) *at $\rho = 2$ that residual reflection ambiguity disappears.*

This is a statement about a reconstruction fiber, not an assertion that a binary degree of freedom is intrinsic to the center itself.

7.1 Two points versus three points

The same distinction between continuous, binary, and unique reconstruction appears already inside one triplet. One observed point known only to belong to some C_3 orbit leaves a continuum of possible centers. Two unordered points known to belong to one equilateral C_3 orbit admit the two classical equilateral completions, hence two possible centroids. A complete three-point orbit has the unique centroid. This observation is elementary, but it provides a minimal local analogue of the boundary-fiber collapse in Theorem 7.2.

Two alternating C_3 orbits: two reflection-conjugate boundaries

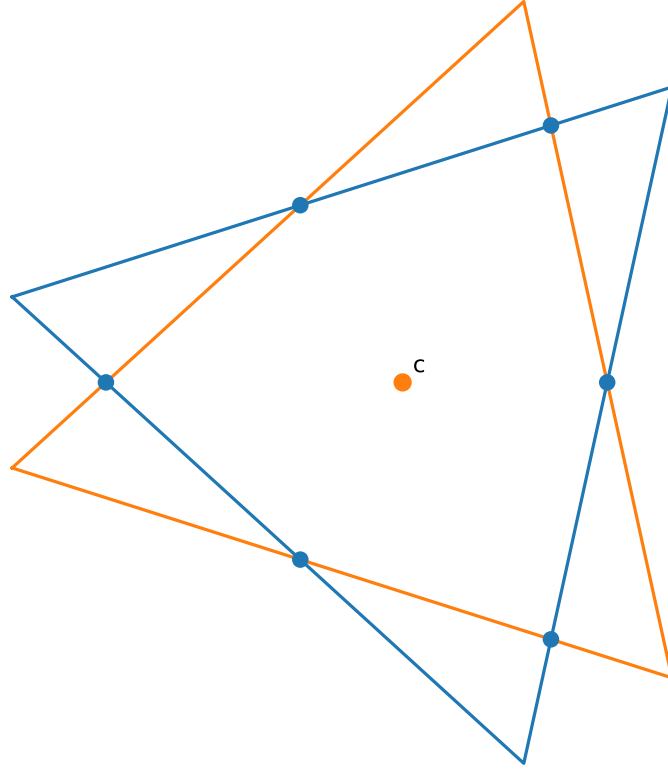


Figure 4: A generic pair of alternating three-point orbits determines two centered equilateral boundaries, related by reflection.

8 Computational verification

The revised verification script regenerates its test objects rather than trusting the supplied CSV tables. The current exact and numerical controls are:

- the shell decomposition is regenerated in integer A_2 coordinates for $1 \leq k \leq 200$, with exact orbit counts and zero orbit sums;
- finite refinements R_{3m} are regenerated for $1 \leq m \leq 80$ and checked against the complete-shell and first-boundary formulas;
- 1000 random one-triplet phases verify the support-function formula in Theorem 5.1;
- 1500 random two-triplet configurations verify (15), (16), and both reflection-conjugate boundary phases;
- all 15 perfect matchings of the six contact points are independently enumerated for 507 deterministic/random cases: exactly two distinct centered equilateral support-line triples survive for $1 \leq \rho < 2$, while the candidates coalesce to one at $\rho = 2$;
- additional controls with $\rho > 2$ confirm the failure of the necessary reduced-offset equation;

- an independent second-round verifier generates 5000 strictly convex C_n -symmetric $2n$ -gons for $3 \leq n \leq 12$ and checks that the even and odd edge orbits each form a regular n -gon support family, with no additional edge perfect matching available.

These computations are checks of the analytic formulas, not substitutes for the proofs in Theorems 3.1, 5.2 and 7.2.

9 Relation to standard geometry and novelty scope

The triangular lattice is the lattice of Eisenstein integers up to similarity, with six nearest-neighbor directions; this realization is standard and appears throughout the literature on planar lattice enumeration, including Baake–Scharlau–Zeiner [3]. Likewise, the coordination sequence of A_2 , and therefore the count $|S_k| = 6k$, is classical [1]. The decomposition into C_3 orbits is then an elementary consequence of the free order-three action. These facts are background rather than standalone novelty claims.

The continuous part sits especially close to the theory of rotationally symmetric convex bodies. Cañete–Miori–Segura Gomis [6] construct a smallest equilateral triangle containing an arbitrary 3-rotationally symmetric convex body and observe nonuniqueness for the regular hexagon. Cañete–Schnell–Segura Gomis [7] extend the rotationally symmetric framework to general order k . Those papers optimize a geometric functional and do not classify all regular polygons whose sides collectively contain a prescribed $2n$ -vertex orbit configuration. General minimum-area enclosing triangles are algorithmic optimization problems [8], the fixed-angle variant is again an optimization problem [9], and tripod configurations concern concurrent normal/support geometry [5]. Our distinction is therefore the exact vertex-incidence fiber and its two-perfect-matching mechanism, not the existence of a rotationally symmetric enclosing polygon.

The contribution claimed here is therefore the following package:

- (i) the canonical C_3 orbit family on S_k gives $2k$ disjoint centroid certificates and transversal number $2k$;
- (ii) one embedded certificate has a complete centered boundary fiber homeomorphic to S^1 ;
- (iii) every strictly convex C_n -symmetric $2n$ -gon has exactly two regular n -gons incident to all its vertices, corresponding to the two perfect matchings of its edge cycle;
- (iv) in the D_3 radial family, the two generic triangles are congruent reflection conjugates with the explicit scale (16), and they coalesce exactly at the midpoint–vertex degeneration $\rho = 2$.

We have not identified this exact vertex-incidence formulation in the sources examined. Because the proof is elementary once the correct edge-cycle viewpoint is introduced, we make no broad priority claim and treat the neighboring rotational-enclosure literature explicitly.

10 Discussion

The discrete and continuous results share one mechanism: a 120° orbit annihilates its displacement from the center under averaging. In the lattice shell, this produces many disjoint redundant center certificates. In the boundary problem, however, center reconstruction is much weaker than boundary reconstruction. A single triplet determines the center while leaving one continuous parameter; the alternating partner removes that continuous ambiguity but leaves a reflection pair.

The general theorem already extends the boundary-incidence mechanism from triangles to regular n -gons. Natural next problems are therefore genuinely beyond the present scope: classify non-strict C_n -symmetric degenerations in which alternating support families may coalesce, treat partial vertex-incidence constraints, analyze more than two rotational vertex orbits, and quantify stability when the prescribed contacts are perturbed. On the discrete side, one may also classify reconstruction thresholds when incomplete canonical orbits are allowed and the center must be recovered from larger subsets.

Data and code availability

The supplementary archive contains the exact shell tables, boundary sweeps, random similarity controls, the manuscript source, and independent verification scripts used in Section 8. The archival reproducibility package is assigned Zenodo DOI 10.5281/zenodo.22003771.

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